Western Mathematics

TOPIC REVISION
QUESTIONS

Mathematics Standard 2

MS-S5: The Normal Distribution

Section I –Multiple Choice Questions (1 Mark each)

Section II – Longer Questions (Marks indicated beside questions)

Section I

10 marks

1. A group of 120 workers took a competency test in welding and their results were normally distributed. The mean score of the group was 90% with a standard deviation of 5%.

   Kevin scored 80% on the test. What is this as a z-score?

   (A) –2

   (B) –1

   (C) 1

   (D) 2

2. The pulse rates (in beats per minute) of athletes at a camp are normally distributed.
   The mean rate is 62 bpm and the standard deviation is 4.5 bpm.
   What percentage of athletes will have a pulse rate lower than 71 bpm?

   (A) 68%

   (B) 81.5%

   (C) 95%

   (D) 97.5%

3. The ages of the students in Year 12 at Alderton College are normally distributed.
   The mean age is 17.7 years and the standard deviation is 0.45 years.
   What percentage of students would you expect to be aged between 16.8 years and 17.7 years?

   (A) 34.0%

   (B) 47.5%

   (C) 68.0%

   (D) 81.5%

4. The mass of a component made by a factory are normally distributed.

   The mean mass is 650 g and the standard deviation is 2.4 g.

   What percentage of the components would you expect to have a mass between 642.8g and 654.8?

   (A) 81.50%

   (B) 95.00%

   (C) 97.35%

   (D) 99.70%

5. The heights of the 120 female students in Year 12 at Maryvale High are normally distributed.

   The mean height is 1 750 mm and the standard deviation is 20 mm.

   How many students would you expect to have a height between 1 710 mm and 1 770 mm?

   (A) 82

   (B) 90

   (C) 98

   (D) 114

6. A survey was done of the incomes of 2 000 full time employed adults. The incomes were normally distributed with a mean of $1 516 with a standard deviation of $145.

   Approximately how many individuals from those surveyed had an income below $1 226?

   (A) 50

   (B) 100

   (C) 200

   (D) 400

7. On a statistics test, the mean was 75 with a standard deviation of 8.

The teacher gave the students marks back as z-scores and Kaylee scored 2.

What was Kaylee's actual raw mark?

(A) 77

(B) 83

(C) 87

(D) 91

8. Angela does two exams in the one day and scores 55 on her Science exam and 48 on her Maths exam.

The Science exam had a mean of 45 with a standard deviation of 5.

The Maths exam had a mean of 42 with a standard deviation of 3.

If the results on both exams are normally distributed, on which exam did she score better, relative to the rest of the cohort?

(A) Both the Maths exam and the Science exam results were equal

(B) The Maths exam result was better

(C) The Science exam result was better

(D) There is insufficient information to compare the two

9. Dinah and Will both play the game *World of Peacecraft* in separate online tournaments.

Dinah scores 12 200 on her tournament which had a mean of 11 450 with a standard deviation of 500.

Will's tournament had a mean of 10 320 with a standard deviation of 350.

What score would Will need to achieve on his tournament to have equal merit with Dinah?

(A) 9 950

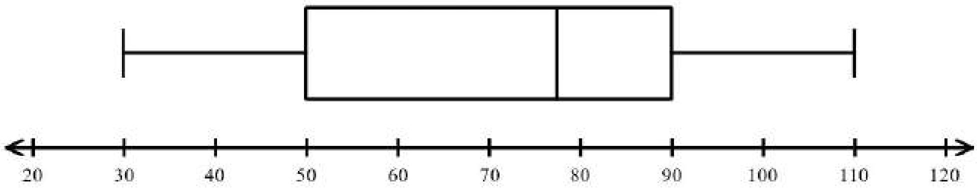
(B) 10 670

(C) 10 760

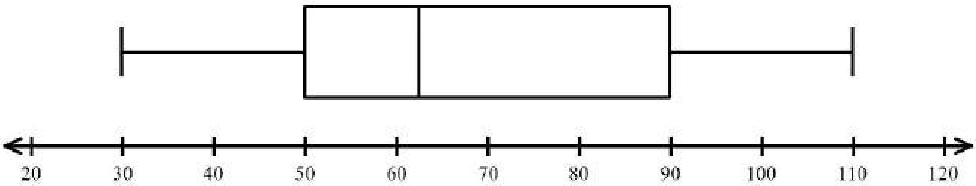
(D) 10 845

10. Which of these box plots could represent a normal distribution?

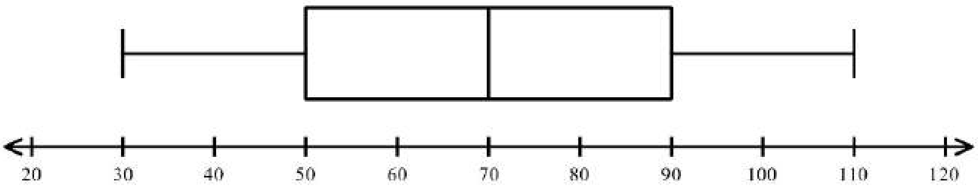
(A)



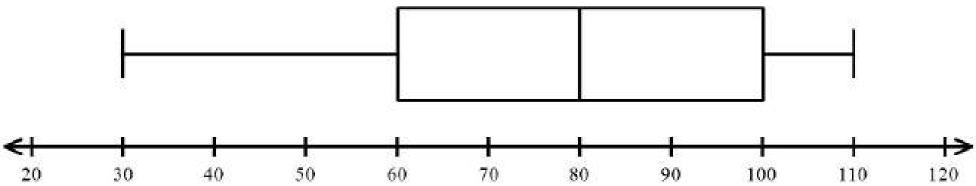
(B)



(C)



(D)



Western Mathematics
Mathematics Standard 2
MS-S5: The Normal Distribution
TOPIC REVISION QUESTIONS - Section II

Instructions
- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (4 marks) **Marks**

Two friends sit the same test at school. The test has a mean mark of 58.0 with a standard deviation of 8.0.

The marks were normally distributed.

(a) Brian gets a mark of 64. What is his z-score? **1**

………………………………………………………………………………………………

………………………………………………………………………………………………

(b) Ahmed gets a z score of -0·5. What is his mark on the test? **1**

………………………………………………………………………………………………

………………………………………………………………………………………………

(c) What percentage of the students will score marks between 50 and 66? **1**

………………………………………………………………………………………………

………………………………………………………………………………………………

(d) Fifty students sat the test altogether. **1**
How many of them would be expected to have achieved a mark greater than 66?

………………………………………………………………………………………………

………………………………………………………………………………………………

Question 12 (5 marks)

The scores of 3 students on their History and Geography assessments are given below, along with the mean and standard deviation of each assessment. Both sets of scores were normally distributed.

Student	History	Geography	
Ally	72	67	
Bella	66	72	
Catalina	85	63	
Mean	60	55	
Standard Deviation	12	8	

(a) Convert Ally's History and Geography marks to z - scores and determine which result was better in comparison to the other students in the course. **1**

……………………………………………………………………………………………

……………………………………………………………………………………………

(b) What percentage of students scored higher marks than Catalina on the Geography assessment? **2**

……………………………………………………………………………………………

……………………………………………………………………………………………

……………………………………………………………………………………………

……………………………………………………………………………………………

(c) The top 2.5% of students on each assessment are invited to a talented students' day. Which students would be invited, and on which result would their invitation be based? Justify your answer mathematically. **2**

……………………………………………………………………………………………

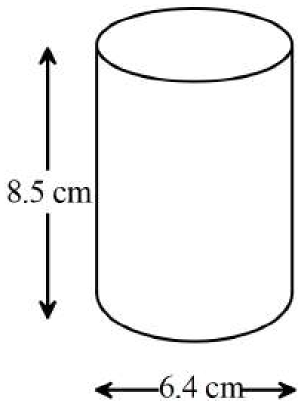
……………………………………………………………………………………………

……………………………………………………………………………………………

……………………………………………………………………………………………

Question 13 (3 marks)

A machine on a production line produces solid metal cylindrical parts. The design specifies a diameter of 6.4 cm and a height of 8.5 cm.



(a) Calculate the surface area of the cylinder. **1**

…………………………………………………………………………………………...

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

(b) In practice the cylinders which come off the production line have diameters which are normally distributed with a mean of 6.3 cm and a standard deviation of 0.05 cm. Any cylinder with a diameter greater than 6.4 cm or less than 6.25 cm must be rejected.
What percentage of the cylinders are rejected? **2**

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

Question 14 (4 marks)

Sally and Heather competed in a trivia competition and a dance competition,

The scores on both competitions were normally distributed.

The mean score in the trivia competition was 70 points with a standard deviation of 12.5, and in the dance competition the mean was 6.2 with a standard deviation of 1.2

(a) Sally scored 95 on the trivia competition. What is this as a z – score? **1**

……………………………………………………………………………………………

……………………………………………………………………………………………

……………………………………………………………………………………………

(b) Heather scored 86 on the trivia competition and 8.6 on the dance competition. In which competition did she do better, compared to the other contestants? **2**

……………………………………………………………………………………………

……………………………………………………………………………………………

……………………………………………………………………………………………

(c) Sally had a z – score of -1.5 on the dance competition. What was her actual score? **1**

……………………………………………………………………………………………

……………………………………………………………………………………………

……………………………………………………………………………………………

Question 15 (4 marks)

Two brands of battery are tested to compare their useful lifetime.

It is found that their lifetimes are normally distributed, with the results below.

Lifetime of Batteries (hours)

Brand	Mean	Standard Deviation
Invigorator	5.4	0.2
Tough Cell	5.9	0.3

(a) John times a Tough Cell battery and finds it lifetime is 6.5 hours. **1**
What is this as a z-score?

……………………………………………………………………………………………

……………………………………………………………………………………………

……………………………………………………………………………………………

(b) An Invigorator battery has a z-score of –1. **1**
What was its lifetime in hours?

……………………………………………………………………………………………

……………………………………………………………………………………………

……………………………………………………………………………………………

(c) A battery is deemed as defective if it has a lifetime of less than 5.0 hours. **2**
Which brand would have the smaller percentage of defective batteries? (Give the percentages.)

……………………………………………………………………………………………

……………………………………………………………………………………………

……………………………………………………………………………………………

……………………………………………………………………………………………

Question 16 (4 marks)

The marks in a class test are normally distributed. The mean is 100 and the standard deviation is 10.

(a) Jason's mark is 105. What is his *z*-score? **1**

……………………………………………………………………………………………

……………………………………………………………………………………………

(b) Mary has a *z*-score of –1.5. What mark did she achieve in the test? **1**

……………………………………………………………………………………………

……………………………………………………………………………………………

……………………………………………………………………………………………

(c) What percentage of marks lie between 90 and 120? **2**

……………………………………………………………………………………………

……………………………………………………………………………………………

……………………………………………………………………………………………

……………………………………………………………………………………………

End of Revision

Western Mathematics MS-S5: The Normal Distribution Revision

Section I - Worked Solutions

No	Working	Answer
1	80% as a z score = $\frac{80-90}{5} = -2.$	A
2	$z = \frac{x-\mu}{\sigma}$ $z_{71} = \frac{71-62}{4.5} = 2$ Approx 50% of z − scores will be less than 0 Approx 95% of z scores between -2 and 2 so 47.5% between 0 and 2 So $47.5 + 50 = \mathbf{97.5}\%$ of z − scores are less than 2	D
3	$z_{16.8} = \frac{16.8 - 17.7}{0.45} = -2$ $z_{17.7} = \frac{17.7 - 17.7}{0.45} = 0$ Approx 95% of z scores between -2 and 2 So $\frac{95}{2} = \mathbf{47.5}\%$ of z scores between -2 and 0	B
4	$z_{642.8} = \frac{642.8 - 650}{2.4} = -3$ $z_{654.8} = \frac{654.8 - 650}{2.4} = 2$ Approx 99.7% of z scores between -3 and 3 so 49.85% between -3 and 0 Approx 95% of z scores between -2 and 2 so 47.5% between 0 and 2 So $47.5 + 49.85 = \mathbf{97.35}\%$ of z scores between -3 and 2	C
5	$z_{1710} = \frac{1710 - 1750}{20} = -2$ $z_{1770} = \frac{1770 - 1750}{20} = 1$ Approx 95% of z scores between -2 and 2 so 47.5 between -2 and 0 Approx 68% of z scores between -1 and 1 so 34% between 0 and 1 So $47.5 + 34 = 81.5\%$ of z scores between -2 and 1 81.5% of 120 = 97.8 = 98 students	C
6	$1516 - 2 \times 145 = 1226$ So 2 SD's below the mean. 95% of scores between $z = \pm 2$ so 5% outside $z = \pm 2$ hence 2.5% below $z = \pm 2$ 2.5% of 2000 = 50	A
7	$z = \frac{x-\mu}{\sigma}$ $\frac{x-75}{8} = 2$ $x - 75 = 16$ $x = 16 + 75 = 91$	D

8	$z = \dfrac{x-\mu}{\sigma}$ $z_S = \dfrac{55-45}{5} = 2$ $z_M = \dfrac{48-42}{3} = 2$ Both the Maths Test and The Science Test were equal	**A**
9	$z = \dfrac{x-\mu}{\sigma}$ $z_D = \dfrac{12200-11450}{500} = 1.5$ $z_M = \dfrac{x-10320}{350} = 1.5$ $x - 10320 = 1.5 \times 350 = 525$ $x = 525 + 10320$ $= 10\,845$	**D**
10	A normal distribution is symmetric with mean and median approximately equal, so only (C) meets this description.	**C**

	MS-S5: The Normal Distribution Section II Solution	Marks
Question 11		
(a)	$z = \dfrac{x - \overline{x}}{\sigma}$ $= \dfrac{64-58}{8}$ $= 0.75$	1
(b)	$-0.5 = \dfrac{x-58}{8}$ $-4 = x - 58$ $x = 54$	1
(c)	68% lie within one standard deviation.	1
(d)	16% score over 66, which is 8 students.	1
Question 12		
(a)	Geog $z_G = \dfrac{67-55}{8} = 1.5$ Hist $z_H = \dfrac{72-60}{12} = 1$ Her Geography result is better.	1
(b)	Z score = $\dfrac{63-55}{8} = 1$ 68% lie between 1 and -1 34% lie between 0 and 1 16% g1reater than 1	2
(c)	5% lie outside -2 and 2 so 2.5% lie above 2. Hist z score of 2 = $60 + 2 \times 12 = 84$ (1 above) Geog z score of 2 = $55 + 2 \times 8 = 71$ (1 above) 2 students would be invited, Catalina on Hist and Bella on Geog.	2
Question 13		
(a)	(i) $SA = 2\pi r^2 + 2\pi r\, h$ $= 2 \times \pi \times 3.2^2 + 2 \times \pi \times 3.2 \times 8.5$ $= 235 \text{ cm}^2$ (3 sig fig)	1

(b)	(ii) Convert to z-scores. For a diameter of 6.4 $z = \frac{6.4 - 6.3}{0.05}$ $= 2$ For a diameter of 6.25 $z = \frac{6.25 - 6.3}{0.05}$ $= -1$ Approximately 68% of scores have z-scores between –1 and 1 Percentage of z-scores less than -1 $= \frac{100 - 68}{2} = 16\%$ Approximately 95% of scores have z-scores between –2 and 2 Percentage of z scores more than 2 $= \frac{100 - 95}{2} = 2.5\%$ Total percentage rejected = 16 + 2.5 = 18.5%	2
Question 14		
(a)	$z = \frac{x - \mu}{\sigma}$ $= \frac{95 - 70}{12.5}$ $= \frac{25}{12.5}$ $= 2$	1
(b)	Trivia $z = \frac{86 - 70}{12.5}$ $= \frac{16}{12.5}$ $= 1.28$ Dance $z = \frac{8.6 - 6.2}{1.2}$ $= \frac{16}{12.5}$ $= 2.0$ The dance score has a higher z score, so is better in comparison.	2

(c)	$z = \frac{x - \bar{x}}{s}$ $-1.5 = \frac{x - 6.2}{1.2}$ $x - 6.2 = -1.5 \times 1.2$ $x - 6.2 = -1.8$ $x = -1.8 + 6.2$ $= 4.4$ Her actual dance score was 4.4	1
Question 15		
(a)	$z = \frac{x - \mu}{\sigma}$ $z_T = \frac{6.5 - 5.9}{0.3} = 2$ Type equation here.	1
(b)	$z_I = \frac{x - 5.4}{0.2} = -1$ $x - 5.4 = -1 \times 0.2$ $x - 5.4 = -0.2$ $x = -0.2 + 5.4$ $= 5.2$ hours	1
(c)	Find what 5.0 hours is as a z − score for both $z_I = \frac{5.0 - 5.4}{0.2} = -2$ So 95% of scores between -2 and 2 and $\frac{5}{2} = 2.5\%$ less than -2 $z_T = \frac{5.0 - 5.9}{0.3} = -3$ So 99.7% of scores between -3 and 3 and $\frac{0.3}{2} = 0.15\%$ less than -3 So Tough Cell has a smaller percentage of defective batteries	2
Question 16		
(a)	$z = \frac{x - \mu}{\sigma}$ $z_J = \frac{105 - 100}{10} = 0.5$	1
(b)	$z_M = \frac{m - 100}{10} = -1.5$ $m - 100 = -1.5 \times 10$ $m - 100 = -15$ $m = -15 + 100 = 85$	1

(c)	Calculate a z − score for both $z_{90} = \dfrac{90-100}{10} = -1$ So 68% of scores between -1 and 1 and $\dfrac{68}{2} = 34\%$ between -1 and 0 $z_{120} = \dfrac{120-100}{10} = 2$ So 95% of scores between -2 and 2 and $\dfrac{95}{2} = 47.5\%$ between 0 and 2 So required percentage $= 34 + 47.5 = 81.5\%$	2